

AI Text Preprocessing Project Report

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Project Title: AI Text Preprocessing Project

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1. Introduction

This project focuses on **text preprocessing** of IMDb movie reviews for natural language processing (NLP) tasks.

Text preprocessing is an essential step before applying machine learning or AI models because raw text often contains noise, irrelevant data, and inconsistent formatting. Cleaning and standardizing text improves model performance, especially for tasks like **sentiment analysis** or **text classification**.

2. Dataset

The dataset used in this project is a **mixed set of IMDb movie reviews**.

For GitHub submission, a **smaller cleaned version** (cleaned_mixeddata_small.csv) was created to reduce the file size below 25 MB.

Columns in the dataset:

- **rating** : Original movie rating given by the reviewer.
- **review** :Original review text.
- **clean_review** :Text after preprocessing.

3. Preprocessing Steps

The following preprocessing steps were applied to the dataset:

1. **Lowercasing**: All text is converted to lowercase to maintain uniformity.
2. **URL removal**: Any links or URLs in the reviews are removed.
3. **Special characters removal**: Only alphabets and spaces are kept; punctuation, numbers, and symbols are removed.

4. **Stopwords removal:** Common words like "the", "is", "and" that do not add meaning are removed to reduce noise.
5. **Output:** The cleaned text is stored in a new column `clean_review` in the CSV file.

4. Files in the Repository

- **cleaned_mixeddata_small.csv:** Smaller cleaned dataset ready for AI model usage.
- **text_preprocessing.py:** Python code used to clean the full dataset.
- **make_small_csv.py:** Python code to create the smaller CSV for GitHub.
- **README.md:** Project explanation.

5. Usage

- The cleaned dataset can be directly used for AI/ML tasks such as **sentiment analysis** or **text classification**.
- Python code files can be run to reproduce the preprocessing or generate a smaller dataset if needed.

6. Notes

- The preprocessing is implemented in **Python** using **pandas** and **regular expressions**.
- Stopwords are **hardcoded**, so no internet connection is required to run the code.
- This project demonstrates the importance of **data cleaning** in natural language processing pipelines.

Repo:

The screenshot shows the main page of the **AI-Text-Preprocessing-Project** repository. At the top, it indicates the repository is **Public** and has **0** watchers. Below this, there are tabs for **main** (selected), **1 Branch**, and **0 Tags**. A search bar labeled "Go to file" and buttons for **Add file** and **Code** are visible. The commit history shows a recent commit by **rajaashar786** titled "Add files via upload" with commit hash **a30c64e** and timestamp "1 minute ago", containing **4 Commits**. Below the commit history, a table lists the files added in this commit:

File Name	Action	Time
AI Text Preprocessing Project Report.docx	Add files via upload	1 minute ago
README.md	README.md	37 minutes ago
cleaned_mixeddata_small.csv	Add files via upload	48 minutes ago
make_small_csvfile.py	Add files via upload	48 minutes ago
textpreprocessing.py	Add files via upload	48 minutes ago

Below the file list, there is a section for the **README** file, which is currently empty.

Github Readme Screen Shot:

The screenshot shows the **README.md** file in the **AI-Text-Preprocessing-Project** repository. The file is titled **AI Text Preprocessing Project** and is 44 lines long (29 loc) and 1.67 KB in size. The content of the README is as follows:

```
1 AI Text Preprocessing Project
2
3 Project Overview
4
5 This project demonstrates text preprocessing on IMdb movie reviews for natural language processing (NLP) tasks.
6 The goal is to clean raw text data, making it suitable for AI or machine learning models such as sentiment analysis.
7
8 Dataset
9
10 The dataset contains mixed movie reviews from IMdb.
11 For this project, a smaller cleaned dataset (cleaned_mixeddata_small.csv) is provided to keep the file size below 25 MB for GitHub.
12 Each row contains:
13
14 rating + original movie rating
15 review + original review text
16 clean_review + cleaned and preprocessed text
17
18 Preprocessing Steps
19
20
21 1. Lowercasing - All text is converted to lowercase.
22 2. URL removal - Any links are removed from the reviews.
23 3. Special characters removal - Only alphabets and spaces are kept.
24 4. Stopwords removal - Common words like "the", "is", and "and" are removed to reduce noise.
25 5. Output - The cleaned text is saved as a new column clean_review in the CSV file.
26 --
```

Output Of Cleaned Dataset:

[7]:

	review	clean_review
0	Wow. That's all I can say. I was freaked out a...	wow thats say freaked movie loved first two of...
1	The Dark Knight Rises capped off what is proba...	dark knight rises capped probably best comic b...
2	I cannot imagine a "Batman" movie after "The D...	cannot imagine batman movie dark knight rises ...
3	The Dark Knight Rises is the third and final i...	dark knight rises third final installment dire...
4	I came out with slight disappointment after wa...	came slight disappointment watching dkr huge e...

+ Code

+ Markdown

